

Make the model the page.

A heuristic review of the main screen, a step-by-step walkthrough of the five pilot tasks, a prioritised fix list with annotated screenshots and sketches, and a measured palette for the model against its ground.

Build	hyperionproj25.github.io/form-anatomy , live on 7 Sep 2026
Viewports	1440×900, 1366×768, 375×812 (2×), Chrome
Measured	Every size, position and colour in this document was read from the rendered page, not estimated
Not done	Sessions with real students. Section 4 is a walkthrough that stands in for them and says what to watch

Fix these five first

Each one is cheap in your terms (colour, layout, defaults, copy) and each removes a reason a student would give up in the first minute.

- 1 Put the model on a dark slate ground.**
Muscle measures 1.80:1 against today's cream stage and the skeleton 1.06:1, which is why the founder cannot see it. The same render on slate measures 5.66:1 and 9.20:1. Panels stay light.
- 2 Remove the intro banner and the start-panel headline.**
Topbar plus banner cost 230px. On a 900px laptop the structure library begins at pixel 893; on a 1366×768 laptop it never appears, and neither do the camera buttons.
- 3 Quiet the defaults, light the choices.**
Five filled green pills (English names, All regions, Any joint, All layers, Both sides) each mean "no filter". They are the most prominent objects on the screen.
- 4 Put the model's actions on the stage.**
A bottom toolbar: views, Move a joint, Layer (Surface/Deep), zoom. Today "Animate" only exists after a joint chip is clicked in the left panel, and Deep does nothing to the model.

5

Change the selection colour and add a rim.

The green is ΔE 3 from the muscle colour for a protanope: invisible to roughly one male student in twelve. Cyan-teal with a bright rim stays distinct for everyone and from origin blue and insertion amber.

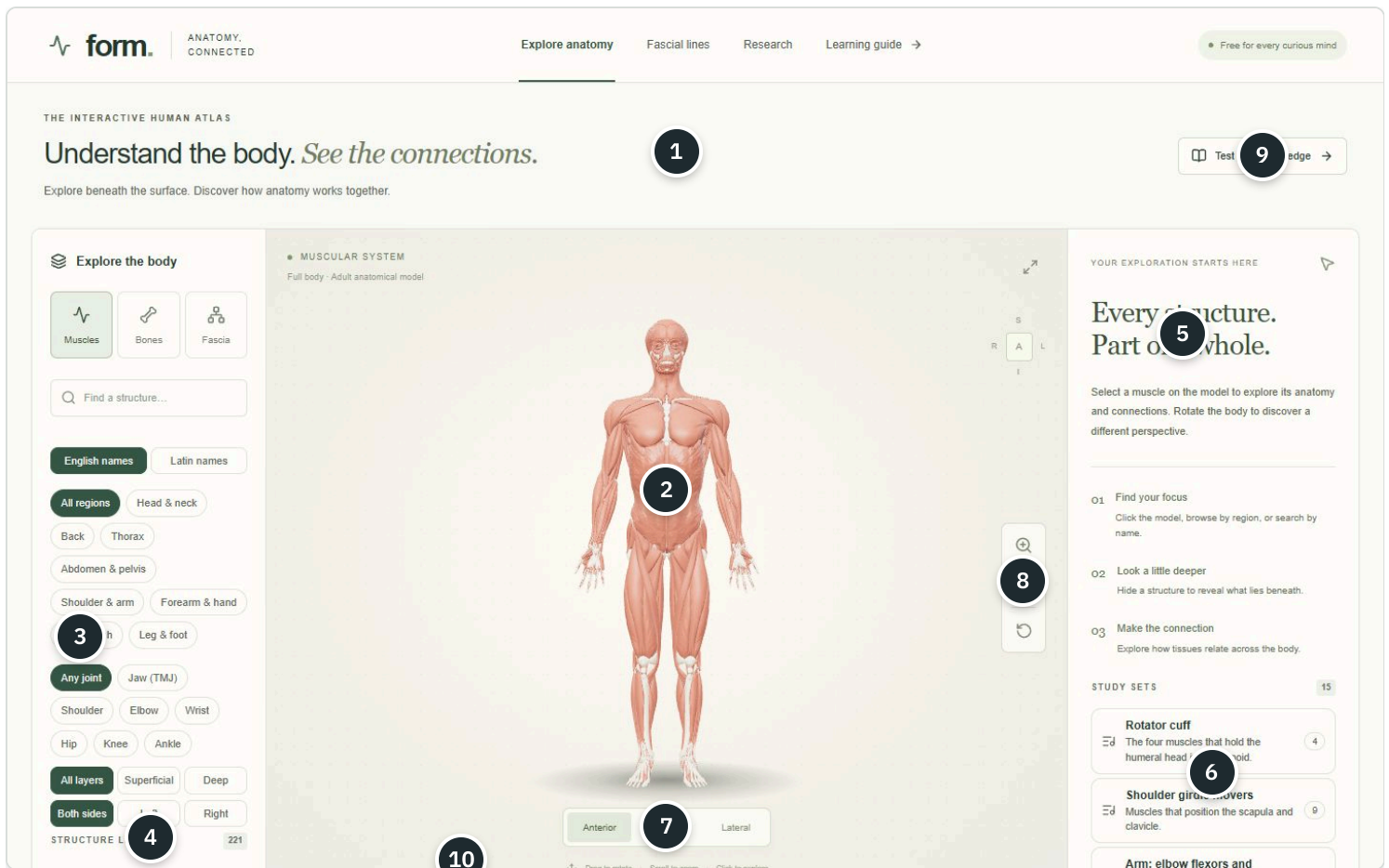
01 How I looked

I read the source to learn what each control does, then drove the live build in headless Chrome to capture every state the pilot guide touches: the start screen, a selected muscle, attachments on, the knee animation, a fascial-line tour, a quiz question, a study set, the handout, the learning guide, and phone width for each. From those renders I sampled the actual on-screen colours of muscle, bone, the selection and the origin and insertion highlights, and computed WCAG contrast and CIE ΔE colour distance, including simulated red-green colour-blindness.

On the usability sessions. I could not sit students down for twenty minutes each. Section 4 walks the five pilot-guide tasks step by step on the real build and predicts where people will hesitate, with the measurement behind each prediction. Treat it as the hypothesis list for your sessions. Your pilot guide already has the right protocol; I have added a "watch for" column so an observer knows what to look at.

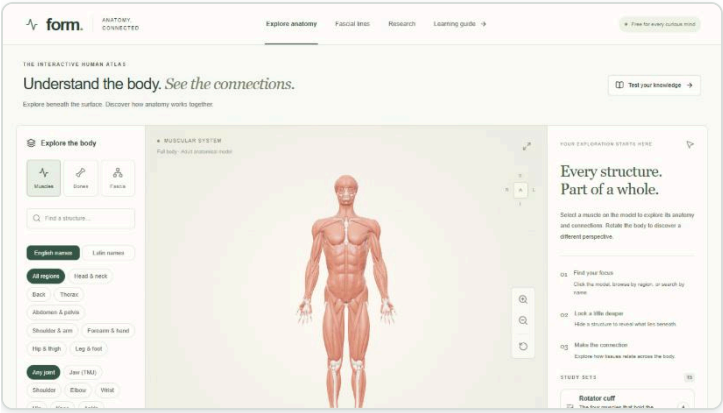
02 The main screen, annotated

This is the first thing a student sees on a 1440×900 laptop after the model loads.

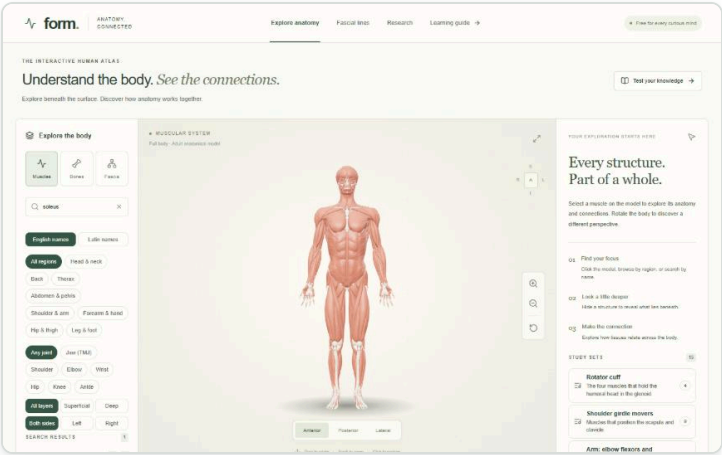


- 1 The banner. 152px of tagline under a topbar that already says "Anatomy, connected".
- 2 The model: about 200px wide on an 838px stage, pastel pink on cream, no edge to speak of. Muscle 1.80:1, bone 1.06:1.
- 3 Five filled pills, each meaning "nothing chosen". The eye goes here first and finds no decision to make.
- 4 The structure library (221 entries) begins at pixel 893. Typing "soleus" in the search box changes nothing a student can see.
- 5 The right panel opens with a headline and three tips. A student looking for what to do reads copy.
- 6 Study sets, the second pilot task, show one and a half above the fold.
- 7 Three camera views are the only visible model controls. Nothing says a joint can move or a layer can be peeled.
- 8 Orientation and zoom float mid-stage, 9px labels.
- 9 "Test your knowledge" is the page's one strong action and it lives in the banner that should go.
- 10 The interaction hint is 8px at 2.4:1. It is also the only place that says "click to explore".

MEASUREMENT	VALUE	WHY IT MATTERS
Space above the workspace (topbar + banner)	230 px	A quarter of a 900px screen, a third of 768px
Top edge of the structure library	893 px	Below the fold on both laptop sizes tested
Model width on the stage	≈200 / 838 px	The subject occupies a quarter of its own stage
Muscle vs stage background	1.80 : 1	Founder's "hard to see" measured
Bone vs stage background	1.06 : 1	Bones mode is nearly invisible (figure in section 6)
Smallest text	8 px	Hints, tags and legend; seven text colours fail 4.5:1
Filter buttons before the list	25 in 5 rows	The list is the thing people came for
Right panel content when a muscle is selected	1472 px in 698	"Show on model" sits 460px below the fold
Phone: header before the stage	295 px	Details sit 1,734px below the model with no cue



1366×768, the commonest student laptop. The library, the camera buttons and the interaction hint are all off screen; the first study set is cut in half.



After typing "soleus". The match exists (Soleus Muscle, L and R) but its row is at pixel 893. The student sees no change and assumes search is broken.

03 Findings, in the order to fix them

P0 items change whether a student can use the tool at all in exam week. P1 items remove friction on the pilot tasks. P2 items are polish and safety nets. Cost uses your own scale: *cheap* is colour, spacing, copy or defaults; *medium* is a layout change; *engineering* touches the viewer.

P0 ·
1

Give the model a dark ground and keep the chrome light

EVIDENCE	Muscle 1.80:1 and bone 1.06:1 against the cream stage; the dot grid adds noise at the same lightness as the model. On the slate ground in section 6 the same render measures 5.66:1 and 9.20:1, and the dimmed state used by attachments and quizzes (currently 1.36:1) becomes readable.
FIX	Stage background <code>radial-gradient(ellipse at 50% 40%, #3e4d55, #2c3940 70%, #232d33)</code> ; remove the dot pattern; stage text <code>#dbe2dc</code> , stage muted <code>#aab4ad</code> . Cards that float over the stage stay paper-white so they read as UI, not anatomy. Lighter mid-grey grounds were tested and rejected: muscle 2.0:1, muddy.
COST	cheap · one CSS block, three text tokens, the ground-shadow disc turned down

P0 ·
2

Remove the banner and the start-panel headline; let the workspace fill the viewport

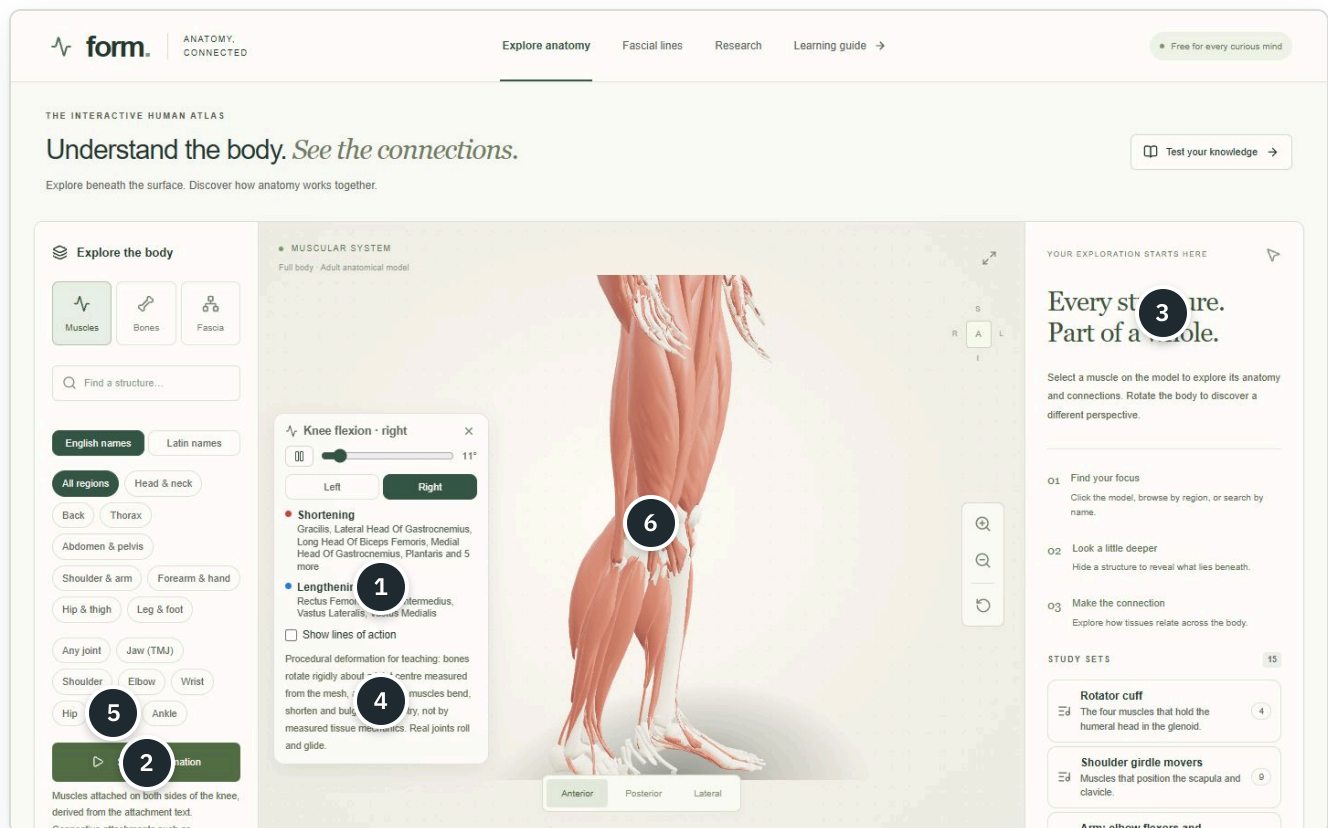
EVIDENCE	230px above the workspace; library at 893px; at 1366×768 the camera buttons sit at 836px. The workspace is fixed at 700px tall, so a bigger screen gets more page below, not more model.
FIX	Delete the intro block. "Test your knowledge" becomes "Quiz" in the top nav. Workspace height <code>calc(100vh - topbar)</code> with the panels scrolling inside. The right panel's start state becomes a short "Start here" list (section 5) instead of "Every structure. Part of a whole." Drop the "Free for every curious mind" badge and the footer tagline; put "Free, no account" in the About dialog.
COST	cheap

Invert the filter emphasis and halve the rows

EVIDENCE	Five dark-filled "All/Any/Both" pills and 25 chips in five rows before the list. Filled means "selected" everywhere else in the UI (view selector, system switch), so the defaults read as choices the student already made.
FIX	Only an active filter is filled; defaults are plain text. Region and Joint become two compact selects on one row (they are lists of 8 and 7). Side stays segmented. Layer moves to the stage toolbar because it now acts on the model. The English/Latin toggle moves to the panel foot next to opacity. Result: search, system switch, one filter row, then the list, all within 260px of the panel top.
COST	cheap

Put "Move a joint" where the model is

- EVIDENCE** The animate button renders only after a joint chip is chosen, at pixel 801 of the left panel; the Learning guide is the only text that mentions it, in a 100-word paragraph. When the knee is animating, the right panel still shows the welcome headline (figure below).
- FIX** A stage toolbar (sketch in section 5): **Anterior** • **Posterior** • **Lateral** | **Move a joint** ▾ | **Layer: Surface** • **Deep** | zoom • reset. "Move a joint" opens a seven-item menu; choosing one runs the existing motionStart, frames the joint and opens the motion card. Every muscle's Crosses row also gets a "Move the knee" link, so a student reading about rectus femoris finds it. Rename "Animate the knee" to "Move the knee": students say move, the app says animate.
- COST** cheap • the actions exist; this is placement and one menu



- 1 The motion card, 236×385px, sits over the model it explains.
- 2 The only way in: a button that appears after clicking a chip, at the bottom of the left panel.
- 3 The right panel is still the welcome text while a knee bends. The panel should show what is happening.
- 4 The caveat is a five-line paragraph. Keep it, shorten it, and give it a shape (P1 · 12).
- 5 "Knee" is one of seven chips in a row labelled by nothing; a student would not guess it animates.
- 6 The camera framing is good. Nothing marks the joint centre or which segment is moving.

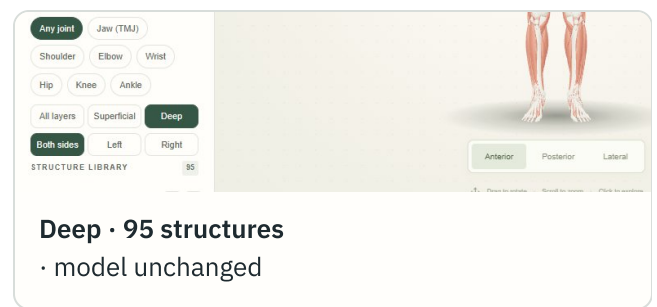
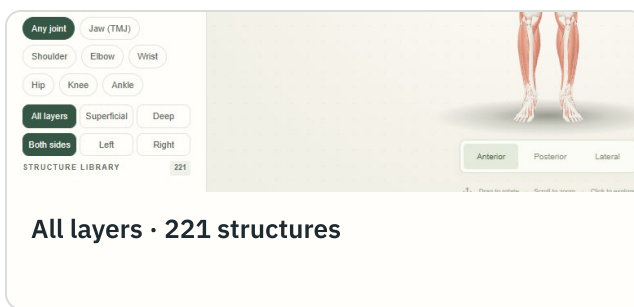
P0 •
5

Make Deep peel the model, not just the list

EVIDENCE With Deep on, the list drops from 221 to 95 entries and the render is pixel-identical (figure). The founder's "the buttons do nothing" is exactly what happens on the model.

FIX In computeStyles: when layer is **Deep**, muscles classified superficial drop to opacity 0.12 and stop catching hover; deep muscles stay full. When **Surface**, the model is unchanged and only the list filters. Show a stage caption while peeled: "Surface layer peeled · approximate". Rename the control **Layer: Surface · Deep** and move it to the stage toolbar. Keep "approximate" in the caption: the classification comes from bounds, and soleus already lands in Deep, which is right.

COST cheap to medium · one branch in a pure function plus a caption



The count changes, the model does not. A student who clicks Deep looks at the model, not at a number.

P0 •
6

Replace the selection green with cyan-teal and a bright rim

EVIDENCE Rendered selection #6bb392 against rendered muscle #dfad9b: ΔE 47 for typical vision, **3 under protanopia, 15 under deuteranopia**. The origin and insertion highlights survive (ΔE 45 to 138) because they differ in lightness as well as hue. Two collisions elsewhere: the compare pins reuse origin blue and insertion orange, and the motion card draws "shortening" cables in red over a red model.

FIX Selection fill #2ac7e0 with the existing emissive, plus the back-face halo brightened to #eafffb and widened so the rim carries the cue for anyone. ΔE against muscle becomes 61 / 33 / 49 (typical / protan / deutan). Pins take violet, magenta, yellow and deep purple. Motion cables use insertion amber for shortening and origin blue for lengthening; the two overlays never coexist, and the metaphor holds: warm is the end that moves. Full table in section 6.

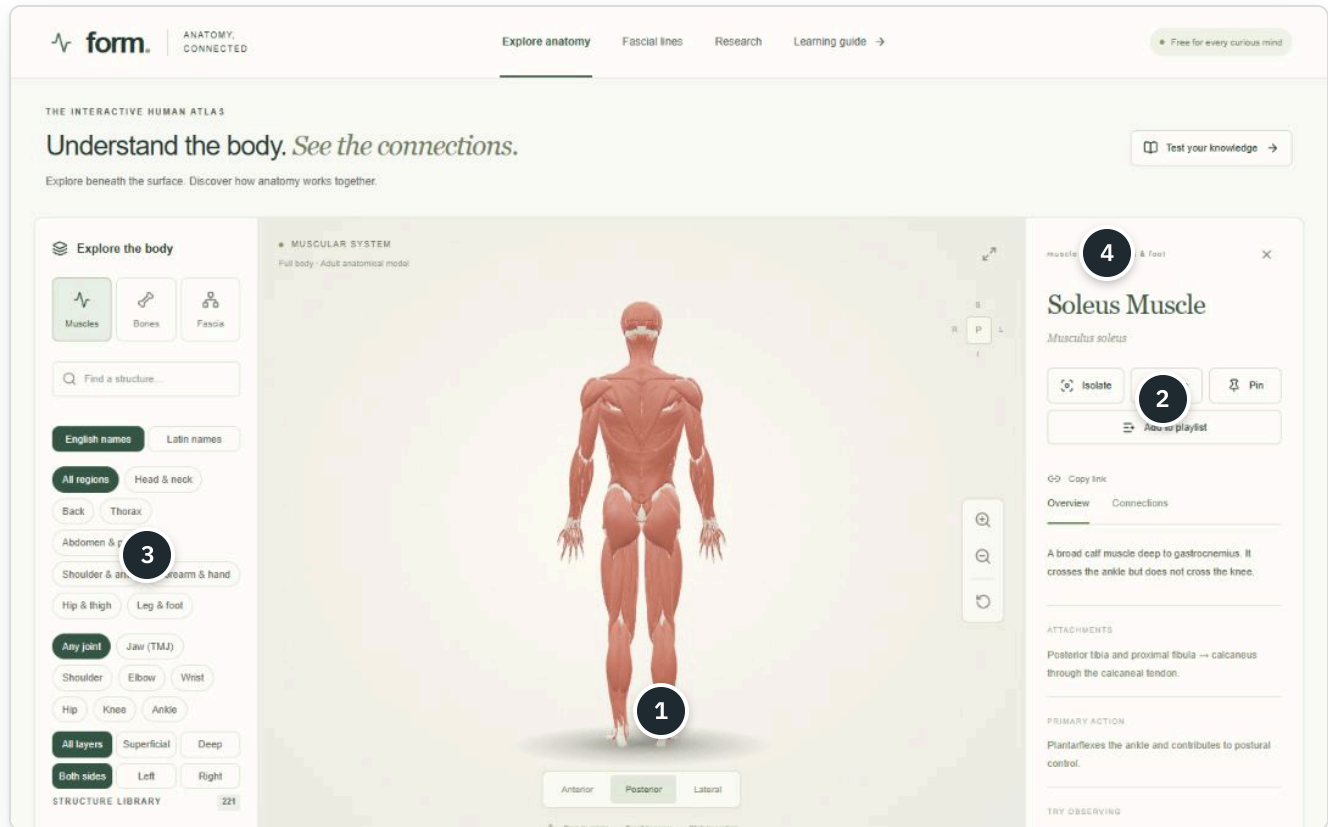
COST cheap · four constants

Give each panel one job and say it in the heading

EVIDENCE Left is headed "Explore the body" and contains a system switch, search, names toggle, three chip rows, an animate button, two segmented rows, the list, opacity, restore, and a help link. Right is headed "Your exploration starts here" and, once a muscle is selected, shows four equal-weight buttons, a copy-link, tabs, prose, facts, attachments and applied notes in a 1472px scroll.

FIX Left heading: **Find a structure**. Right heading: **Selected** with the structure name, or **Nothing selected** with three starting points (study set, quiz, fascial line) and the study-set list. In the detail state, Attachments becomes a chip directly under the name alongside Isolate, Hide, Pin and +Playlist (icon plus label, one row), so the pilot's "Show on model" is visible without scrolling. Facts stay as they are; they are the best part of the panel.

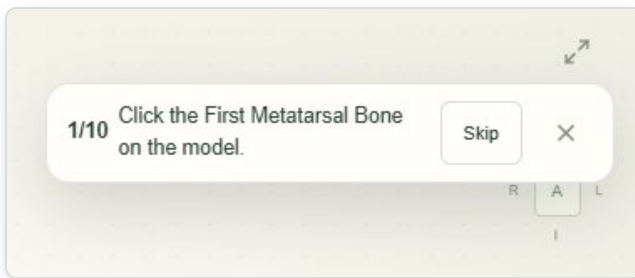
COST cheap



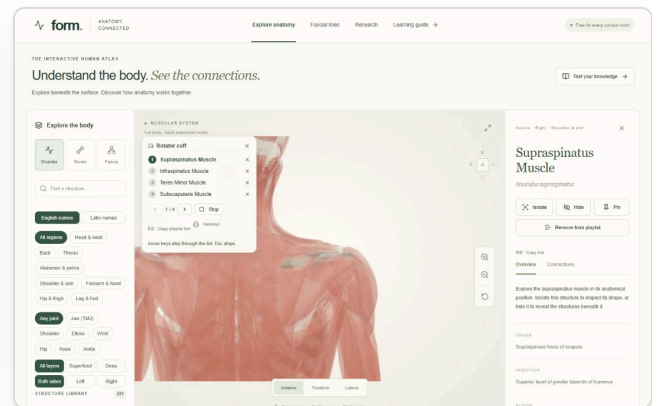
- 1 The selected soleus: a 30px green patch. On slate with a rim it reads at a glance (section 6).
- 2 Four equal buttons. Attachments, the one students need, is 460px lower.
- 3 The list is unchanged because it is below the fold; the selection has no echo on the left.
- 4 The kicker "muscle · Right · Leg & foot" is 8px. It is useful; make it 12px.

One dock for floating cards, and nothing over the orientation gizmo

- EVIDENCE** The quiz card overlaps the orientation widget (crop below). The playlist and motion cards sit top-left and cover the model at 236 to 260px wide. On a phone the motion card covers the whole stage.
- FIX** A single dock at the bottom-left of the stage holds one card at a time (quiz, playlist, motion) with a collapse chevron that leaves a one-line strip ("Knee flexion · 22° ▶"). Orientation and zoom move into a top-right corner stack. On phones the dock is the bottom sheet (P1 · 11).
- COST** medium



Quiz question over the gizmo. "Click the First Metatarsal Bone" on a 200px-wide model also needs a zoom hint, and the question card hides the reset button students would use to recover.



Rotator cuff study set. The card covers the stage, and the fly-to lands inside the shoulder because supraspinatus is under trapezius. Ghost the occluders when a list selection is hidden (P2 · 15).

Set a type scale with a 12px floor and fix seven failing colours

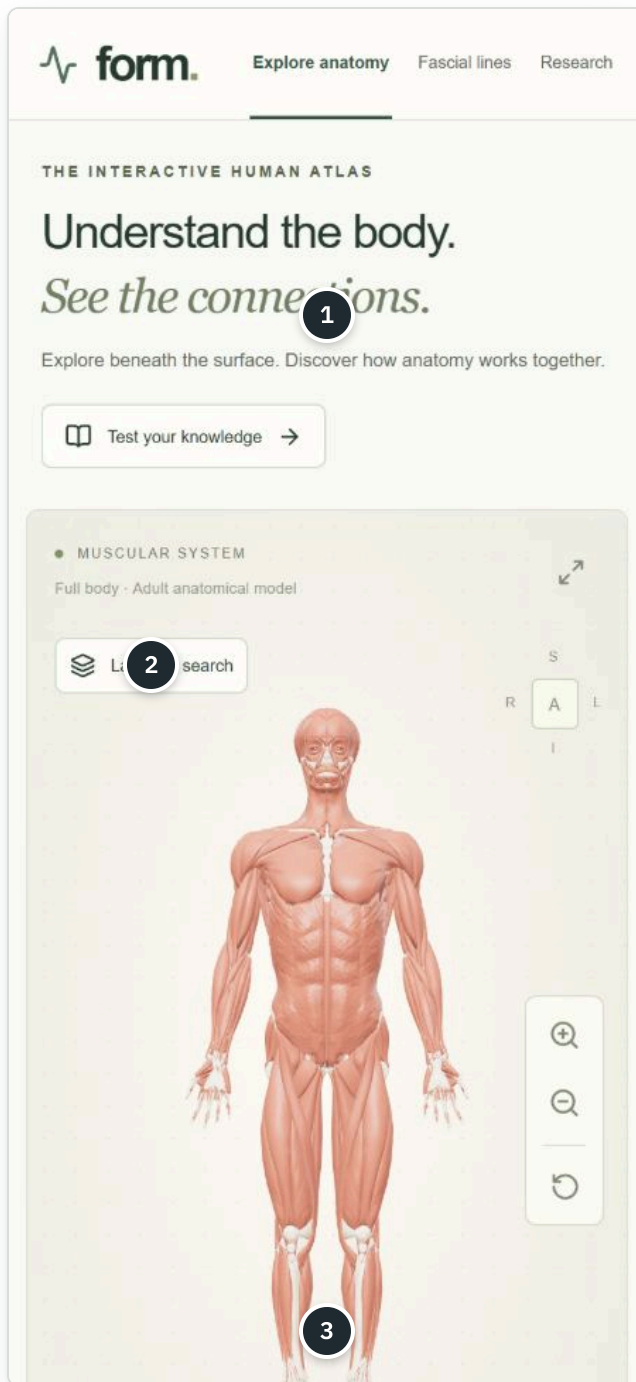
- EVIDENCE** Current sizes 8, 9, 10, 11, 12, 13, 15, 29, 30px. The Lighthouse pass darkened most secondary text, but seven pairs still measure under 4.5:1 (table in section 6), and the automated check cannot catch them: the axe test runs in jsdom with colour-contrast disabled.
- FIX** Scale 12 · 13.5 · 15 · 18 · 24 · 30. Nothing under 12px, including the tags and the stage hint. Secondary text on paper no lighter than #5c665e (5.9:1); on slate no darker than #aab4ad (5.3:1). Keep Georgia for titles if you like the plate feel, but set the Latin name in the same size as body, not 12px italic at 2.65:1.
- COST** cheap

Tell reduced-motion users what is off, and give them the scrubber

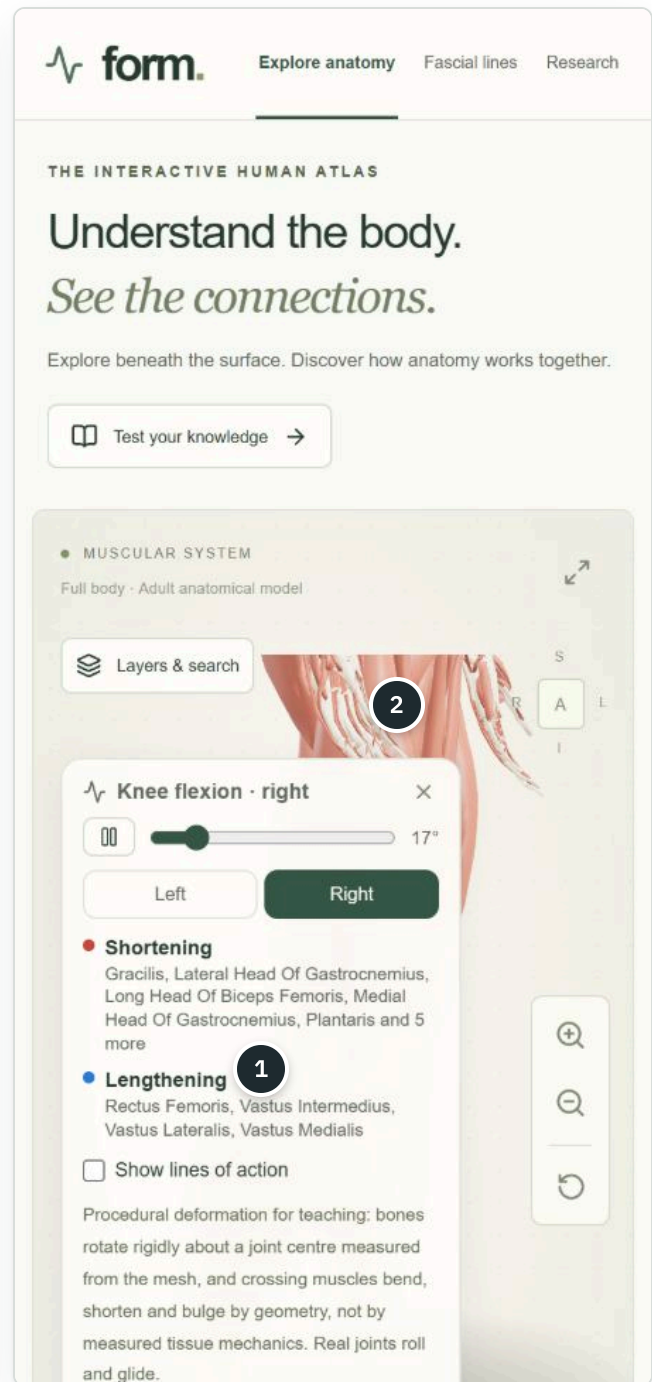
EVIDENCE	With <code>prefers-reduced-motion: reduce</code> , the engine sets <code>motionPlaying = on && !reduced</code> , so Play does nothing: I pressed it and read 0°, then 0° again a second later. The tour's drift and the contraction pulse are also silently off. Snapping instead of tweening is right; a dead button is not.
FIX	Under reduced motion hide Play and show one line in the card: "Animation is off because your system prefers reduced motion. Drag the angle." The tour card says "Auto-advance is off; use Next." Nothing else changes.
COST	cheap

Phone: model first, details in a bottom sheet

EVIDENCE	295px of header before the stage; the stage is a fixed 650px; the detail panel is 1,734px of content below it with nothing telling the student to scroll; the layers drawer covers 80% of the width; the motion card covers the model entirely (figure).
FIX	No banner. Stage height = viewport minus topbar. A search pill floats over the top of the stage and opens the Find panel as a half-height sheet with the list scrolling inside it. Details, quiz, playlist and motion all live in a bottom sheet that peeks at about 96px ("Soleus · Musculus soleus"), drags to half, and expands to full. The stage toolbar shrinks to icons with labels. Sketch in section 5.
COST	medium



- 1 295px of banner and topbar before anything anatomical.
- 2 "Layers & search" is the only control, and it opens a drawer that hides the model.
- 3 The model is cut at the knees; details are another 1,734px down.



- 1 The motion card covers the stage; the caveat alone is 200px.
- 2 The knee that is moving is behind the card.

P1 •
12

Restyle the caveats as a chip that opens, and never hide them

EVIDENCE	The motion caveat is a five-line paragraph inside the card; the fascia legend's caveat ("Teaching path drawn through structure centers. Not a fascial sheet.") is 8px at 2.4:1; the attachments caveat ("Approximate.") is 10px.
FIX	A small labelled chip in the card header, Teaching model for motion and paths, Approximate for attachments, always visible while the overlay is on. Tapping it opens the full sentence; on first use it opens by itself. In the tour caption band the legend sits at 12px in the band's own text colour. This keeps every caveat on screen and readable, and stops it competing with the content it qualifies.
COST	cheap

P2 •
13

Copy pass: name things by what students call them

FIX	• "Explore the body" → "Find a structure". "Your exploration starts here" → "Nothing selected". "Every structure. Part of a whole." → gone. • "Animate the knee" → "Move the knee". "Layer controls" → "Opacity". "A little help exploring" → "Help". • List and card names drop the "Muscle" suffix and title case: "Soleus", "Long head of biceps femoris". The motion card already does this. • "Show on model" → "Show attachments". "Crosses: Ankle" → "Crosses the ankle · Move it". • Evidence pills keep Wilke's terms, and the first badge on any line carries its one-line gloss inline: "Verified · dissection found tissue continuity", rather than in a collapsed disclosure below the fold.
COST	cheap

P2 •
14

Replace the learning guide's wall of text with three anchored hints

EVIDENCE	The guide is five paragraphs of 60 to 100 words in a modal. It is the only place that explains joints, peeling and tours, and nobody opens it before they are stuck.
FIX	On first visit, three small callouts anchored to the UI, one at a time: "Click any muscle" on the model, "Move a joint" on the toolbar, "Follow a line" on the Fascia switch. Dismiss on click; never again. Keep the guide for reference but cut each entry to two sentences.
COST	medium

P2 ·
15

When a selection is hidden, ghost what hides it

EVIDENCE	Selecting supraspinatus from the study set flies the camera into the shoulder; the muscle is under trapezius and deltoid, so the view is a wall of dimmed pink. Soleus from the list is under gastrocnemius from every angle.
FIX	When a selection comes from the list, search, playlist or quiz and its centroid is occluded from the chosen camera, apply the existing "revealed" opacity (0.28) to the occluders until the next selection. The reveal logic already exists for quiz targets.
COST	engineering · an occlusion test and one flag

P2 ·
16

Add a real-browser contrast check so the palette cannot regress

EVIDENCE	The build's axe test runs in jsdom with colour-contrast disabled by necessity. Every failing pair in this review passed CI.
FIX	Run axe once in headless Chrome against the built site (the screenshot driver used for this review does the navigation with no dependencies beyond Node and Chrome), asserting zero colour-contrast violations at 1440×900 and 375×812. I can hand over that script.
COST	cheap

P2 ·
17

Search selects on Enter and shows its results where the eye is

EVIDENCE	Typing "soleus" gives one row with L and R toggles, below the fold. There is no Enter behaviour.
FIX	With P0 · 3 the list is in view. Add Enter to select the first match on the current side, and show a two-line "1 match" strip under the search box that names it. Keep the L/R toggle.
COST	cheap

04

Walkthrough of the five pilot tasks

Each task from docs/pilot-guide.md, stepped through on the live build. "Predicted hesitation" is where I expect a student to stop or go wrong; "watch for" is what an observer

should note. Where the walkthrough could not judge (whether a student understands a word), it says so.

Task 1 · Find the soleus, two ways search, region filter, or clicking the model		
STEP	WHAT THE BUILD DOES	PREDICTED HESITATION · WATCH FOR
Type "soleus" in the search box	Filters the list to one row at pixel 893, below the fold at both laptop sizes	High. The screen does not visibly change. Watch whether they retype, press Enter, or scroll the panel; time to first scroll.
Click the row, choose L or R	Selects; camera does not move; soleus is under gastrocnemius from the front	Moderate. The green patch is 30px and on the back of the leg. Watch whether they rotate to find it or believe nothing happened.
Second way: region "Leg & foot"	List of about 40, also below the fold	Same as step 1.
Third way: click the model	Clicking the calf selects gastrocnemius; the panel copy for gastrocnemius does not mention soleus	High. Watch for repeated clicks on the calf and whether anyone finds "Hide". The soleus lesson text says to hide gastrocnemius, but only after they have found soleus.

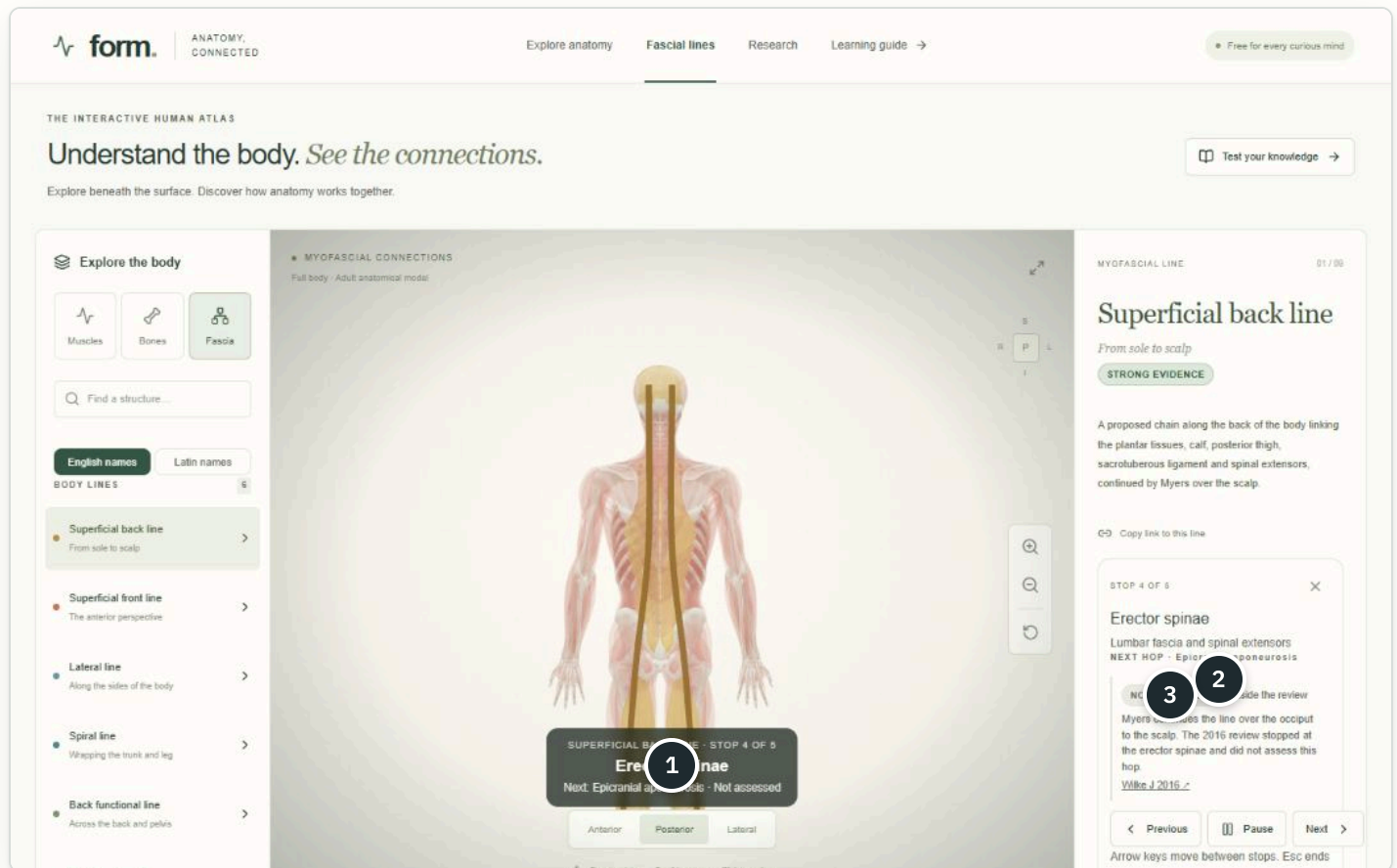
Task 2 · Rotator cuff study set, arrow keys, "Show on model"		
STEP	WHAT THE BUILD DOES	PREDICTED HESITATION · WATCH FOR
Find "Rotator cuff" on the start panel	First study set, at pixel 732; cut in half at 768px	Low at 900px, moderate at 768px. Watch whether they scroll the right panel or the page.
Open it	Playlist card top-left; camera flies into the shoulder; supraspinatus is occluded	Moderate. Watch for "where is it?" and attempts to rotate out of the shoulder.
Step through with arrow keys	Works; the status reads 1/4 etc.; focus must not be in an input	Low once the card's hint is read. Watch whether they click the card first (which is fine) or the stage.
Turn on "Show on model"	Button at pixel 1160 of a 698px panel; label does not say "attachments"	High. Watch how long until they scroll the right panel, and whether they try the left panel's opacity slider first.

Task 3 · Quiz on a region, ten questions, weak spots

STEP	WHAT THE BUILD DOES	PREDICTED HESITATION · WATCH FOR
"Test your knowledge"	Banner button opens a clear modal with region and line grids	Low. The modal is the best-structured screen in the app.
Answer a "click the ... on the model" question	Card top-right over the gizmo; target such as a metatarsal is a few pixels at default zoom; muscles mode hides most bones	High. Watch for zoom attempts, mis-clicks on neighbours, and use of Skip. Ask afterwards whether they knew they could zoom.
Finish ten and read weak spots	Results card with score and "Practice weak spots (n)"; stats also in the quiz modal	Low. The word "weak spots" is plain. Watch whether they expect a list of the missed names, which the card does not show.

Task 4 · Superficial back line tour, read a badge

STEP	WHAT THE BUILD DOES	PREDICTED HESITATION · WATCH FOR
Switch to Fascia	Left panel switch or top nav; both work; SBL is preselected	Low.
Start the tour	"Start the guided tour" is the one strong button in the panel	Low. Good pattern; copy it for Move a joint.
Read one evidence badge	Badge in the tour card and again in the caption band; "How to read the badges" is collapsed below	Cannot judge from a walkthrough. This is the question your sessions must answer. Watch whether "Verified" is read as "this line is real" rather than "dissection found tissue continuity at this hop". Listen for the sentence they produce; if it does not mention dissection or a study, the gloss needs to be inline (P2 · 13).



- 1 The caption band repeats the stop and badge; it is the right place for both, and could take the legend caveat at 12px.
- 2 The tour card repeats it again. One of the two should carry the evidence, the other the controls.
- 3 "Not assessed · outside the review" is readable. Whether it is understood is the session's job.

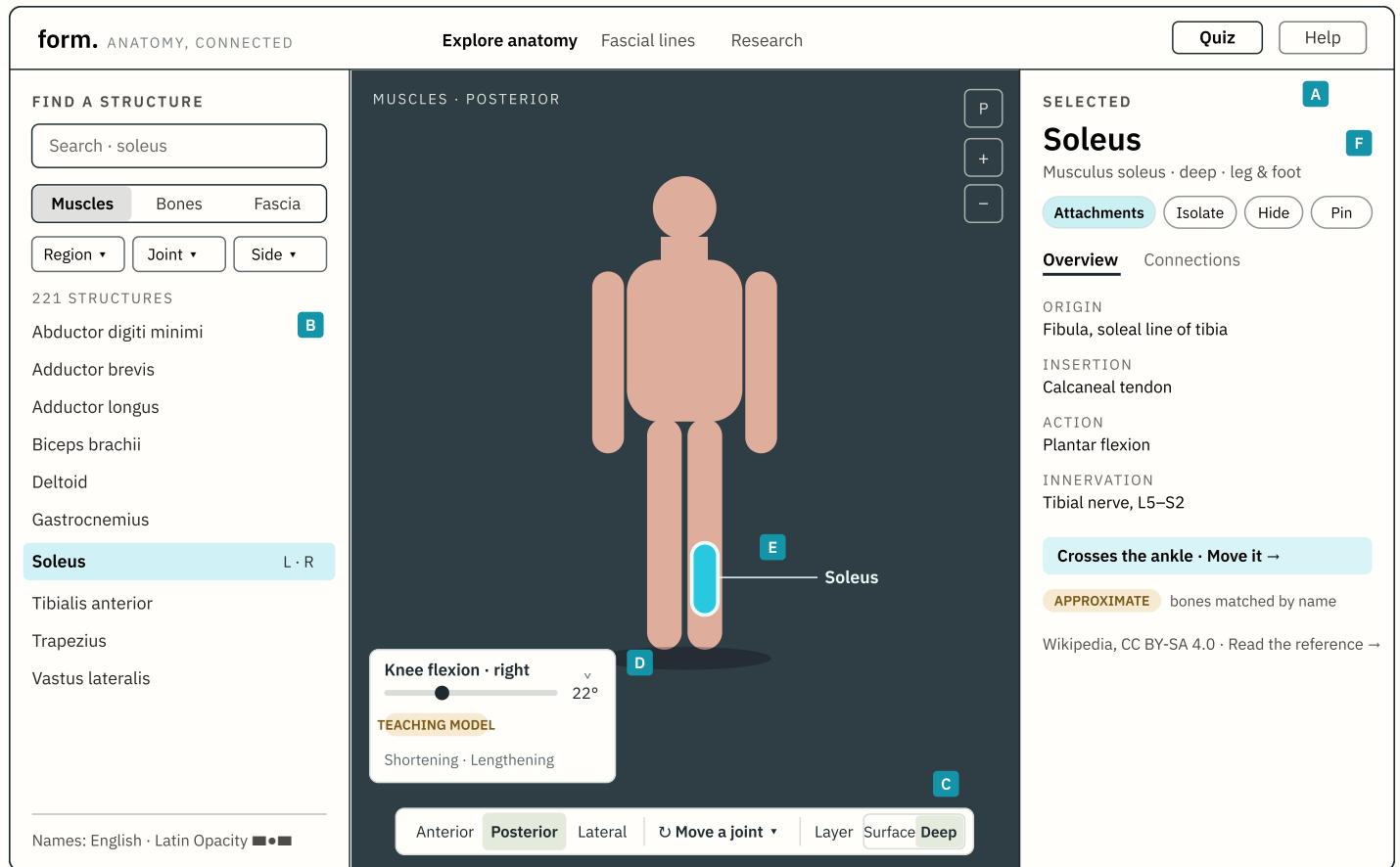
Task 5 · Handout: print or download the Anki deck

STEP	WHAT THE BUILD DOES	PREDICTED HESITATION · WATCH FOR
Find "Handout" on the playlist card	A 12px text button with a printer icon beside "Copy playlist link"	Moderate. Watch whether they look in the right panel first.
Print or download	Modal with a clear title, "Print" and "Download for Anki", and one line of Anki import instructions	Low. The modal is good. Watch only for the browser's download bar on phones.

Two things to add to the after-session questions: "What did you think the Superficial and Deep buttons would do?" and "Did you know the model could move?". Both are the founder's observations; both are cheap to confirm with three people.

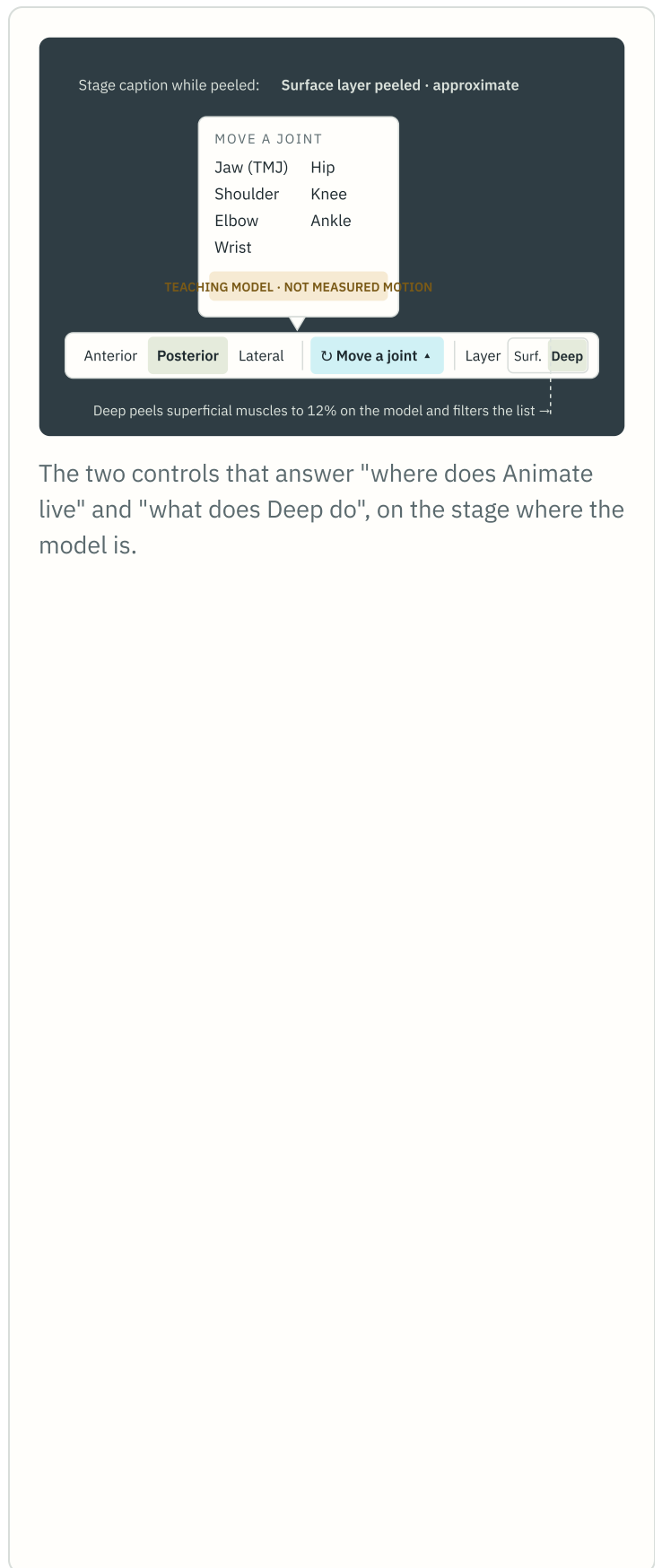
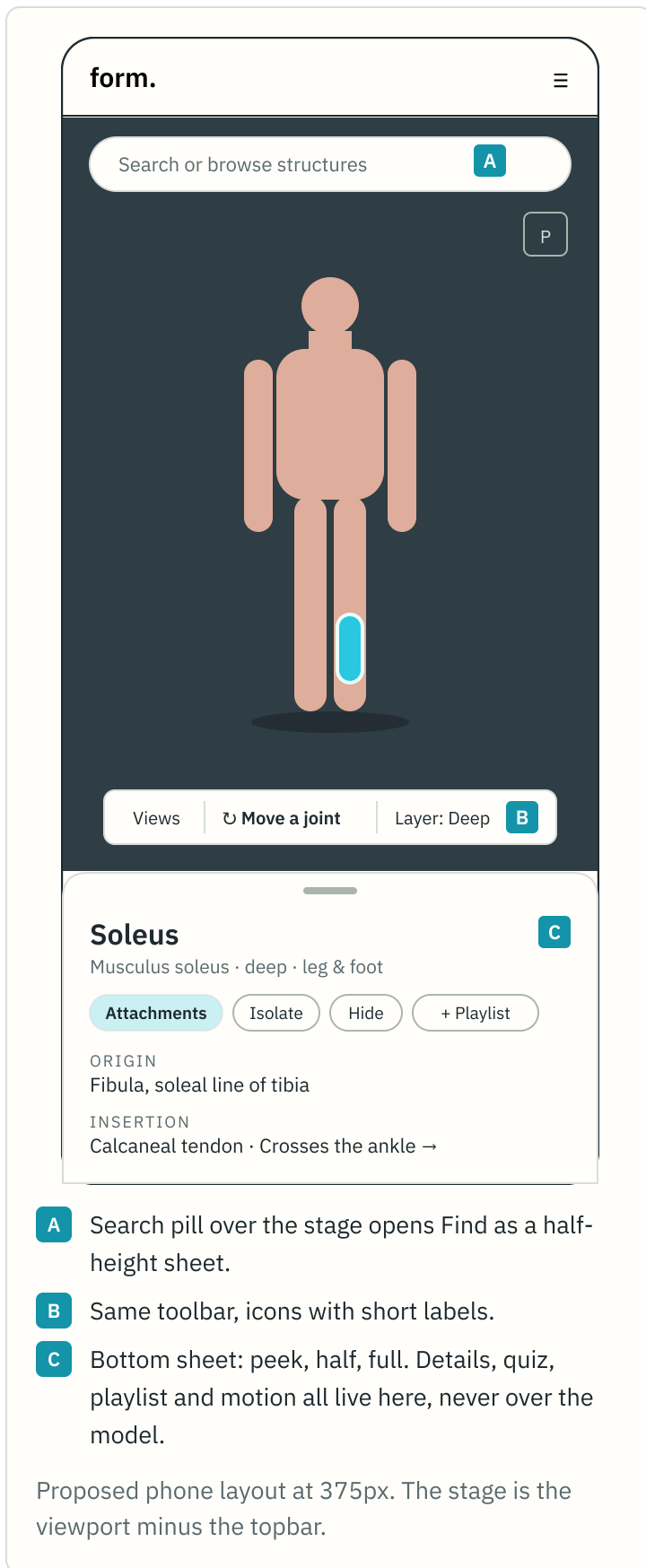
05 Sketches

Rough on purpose. The laptop sketch shows the same three columns with each panel doing one job, the stage carrying the model's own controls, and a dock for transient cards. The phone sketch shows the same content as a bottom sheet. Letters are keyed below each drawing.



- A** No banner; Quiz and Help in the topbar. Workspace fills the viewport.
- B** Search first, one filter row, then the list. Selected row echoes the model.
- C** Stage toolbar: views · Move a joint · Layer (Surface/Deep) · zoom.
- D** One dock for quiz, playlist and motion cards, collapsible to a strip, caveat as a chip.
- E** Selection: cyan fill and a bright rim, with a label leader.
- F** Right panel says what it is; actions are chips under the name; Crosses links to Move.

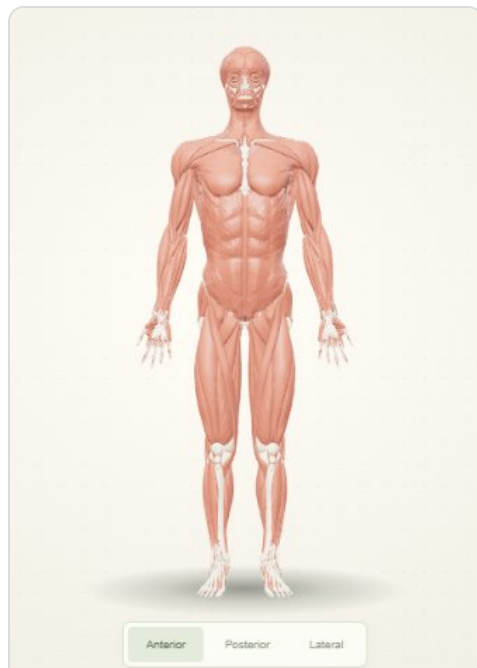
Proposed laptop layout. Same three columns, each with one job; the model is the largest object on the page.



06 Palette and contrast

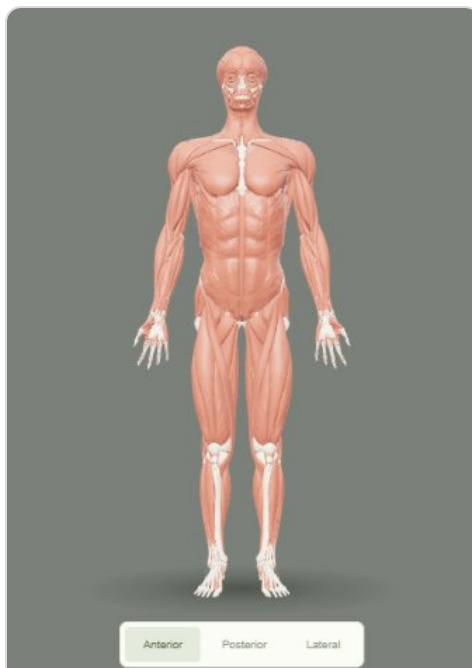
The principle: the model is the only thing on the page allowed to be colourful, so its ground must be dark and neutral, its chrome light and quiet, and its four highlight roles separated by lightness as well as hue so they survive colour-blindness. The renders below are the live model with only the stage CSS changed.

The ground



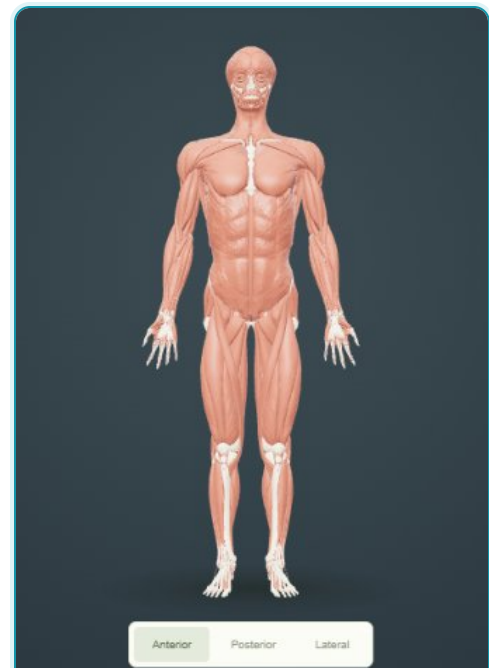
Current cream

#f6f4ec · muscle 1.80:1 · bone 1.06:1



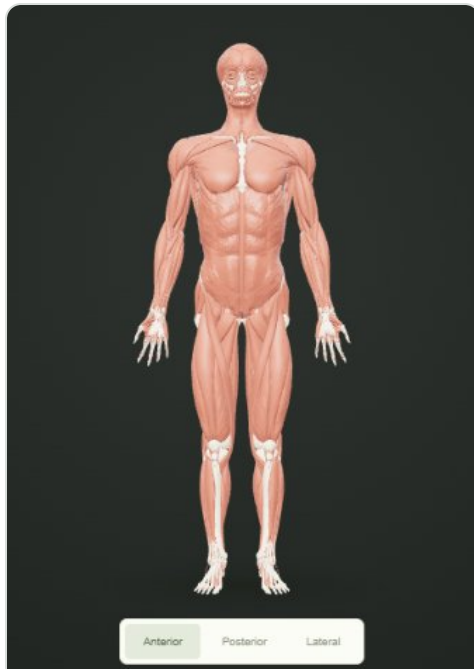
Mid grey, rejected

#7b807a · muscle 2.03:1 · muddy



Slate, recommended

#3e4d55→#232d33 · muscle 5.66:1 · bone 9.20:1



Charcoal, close second

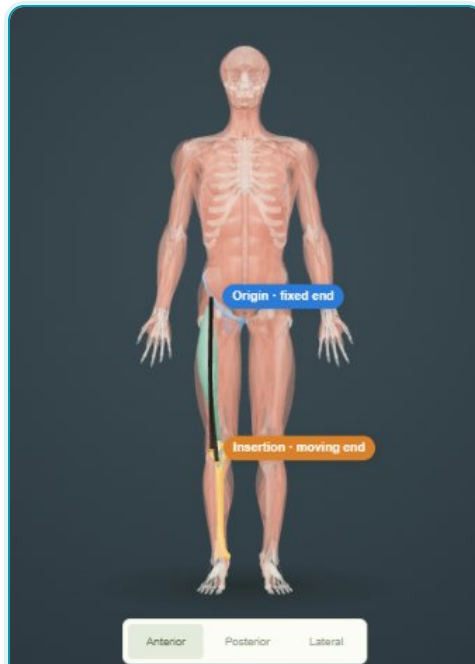
#2e332f → #1b1e1c · heavier,
warmer, less edge on bone

Slate is cool so the warm muscle and bone sit forward by complementary contrast, and it stays a shade lighter than charcoal so the origin blue keeps 3.4:1 against it. The ground-shadow disc should be turned down to about a third of its current opacity on the dark ground.



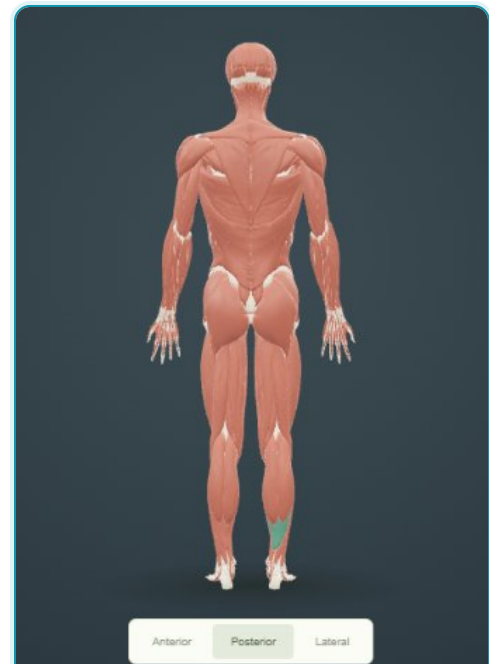
Attachments on cream

dimmed muscles 1.36:1



Attachments on slate

origin, insertion and the
dimmed body all read



Selection on slate

even the current green reads;
cyan plus rim reads for
everyone



Tokens

Stage			Model (material values; highlights carry emissive so they render close to these)		
	Stage gradient, centre	#3e4d55		Muscle, keep (renders ≈ #dfad9b)	#a35b4c
	Stage gradient, mid	#2c3940		Bone, keep (renders ≈ #ece9d9)	#e0d3b7
	Stage gradient, edge	#232d33		Selected fill (was #477965)	#2ac7e0
	Stage text (8.5:1)	#dbe2dc		Selection rim / halo (was #9fc7b4)	#eafffb
	Stage muted text (5.3:1)	#aab4ad		Origin bones (was #2b7bd9, lifted for the dark ground)	#3d8bff
	Cards over the stage	#fffefb		Insertion bones (was #d9822b, toward amber)	#f2a531
Chrome (panels)				Motion: shortening (was red #c8473f)	#f2a531
	Page	#f7f6f2		Motion: lengthening	#3d8bff
	Panel	#fffefb	Compare pins		
	Ink (keep)	#273a31		(no blue, no orange)	#5b3fa6 #a86ee0 #e0569f #e8c547
	Secondary text, floor (5.9:1)	#5c665e			
	Brand green, active states only	#365646			

#dfe3d7 Dimmed structures stay at opacity 0.28; on slate that renders near #8c7f79, 2.9:1 against the ground, which is enough for context and not enough to compete. Fascial line colours can stay; check each against slate once (the sage #809571 and mauve #9e7c9b will want lifting).

The four highlight roles, and what a colour-blind student sees

Rendered colours sampled from the screenshots; simulated with the Viénot–Brettel–Mollon dichromat model. ΔE below about 20 is hard to tell apart on a curved, lit surface.

CURRENT	Ground	Muscle	Bone	Selected	Origin	Insertion
Typical vision	#f6f4ec	#dfad9b	#ece9d9	#6bb392	#2277dd	#dd8822
Protanopia				ΔE 3 vs muscle		
Deutanopia				ΔE 15		
PROPOSED	Ground	Muscle	Bone	Selected	Origin	Insertion
Typical vision	#2f3d44	#dfad9b	#ece9d9	#2ac7e0	#3d8bff	#f2a531
Protanopia				ΔE 33 vs muscle		
Deutanopia				ΔE 49		

PAIR	CURRENT ΔE TYPICAL / PROTAN / DEUTAN	PROPOSED ΔE TYPICAL / PROTAN / DEUTAN
Selected vs muscle	47 / 3 / 15	61 / 33 / 49
Selected vs bone	38 / 23 / 26	46 / 31 / 43
Selected vs origin	83 / 79 / 79	63 / 53 / 48
Selected vs insertion	77 / 44 / 61	103 / 81 / 101
Origin vs insertion	123 / 119 / 138	132 / 130 / 147
Insertion vs muscle	49 / 45 / 50	51 / 49 / 53
Muscle vs ground	1.80:1	5.66:1
Bone vs ground	1.06:1	9.20:1
Origin vs ground	4.01:1	3.38:1

Text that fails today, and what to set it to

ELEMENT	NOW	RATIO	SET TO	RATIO
Interaction hint	8px #97a08a on stage	2.41	12px #c2cbc4 on slate	6.75
Stage subtitle	9px #949d8b	2.49	12px #aab4ad on slate	5.26
Line legend and its caveat	9px / 8px #7b846f #9aa18e	3.45 / 2.36	12px #e6ebe4 in the caption band	9.28
View selector, inactive	10px #859077	3.25	12px #5c665e on paper	5.8
Latin name	12px italic #9aa18e	2.65	13.5px #5c665e	5.92
Line chapter "01 / 09"	9px #a4ac99	2.33	12px #5c665e	5.92
"Free for every curious mind"	10px #6e7b64	4.02	Remove	—
Icon buttons	#7f8a76	3.20	#5c665e (icons pass at 3:1, but weak)	6.0

Evidence pills, section labels, body copy, the detail title and the active chips all pass today and can keep their colours. The badges stay on light paper in the panel where they are, at 12px, and they keep Wilke's wording.

07 The five questions

Two panels or one flow?

Two panels on a laptop, one flow on a phone.

The split is right; the jobs are wrong. Left is where you *find* things, right is where you *read about the one thing you chose*. Make each heading say that, move everything that acts on the model (opacity, layer, animate) out of both and onto the stage, and keep the right panel's empty state to three starting points and the study sets. On a phone the same two jobs become a search sheet and a details sheet over a full-height stage.

Where should "Animate this joint" live?

On the stage toolbar as "Move a joint", plus a link on every muscle that crosses one.

It is an action on the model, so it belongs with the camera controls, visible before any filter is chosen. A menu of the seven joints is faster than the chip row and needs no explanation. The second entry point, "Crosses the knee · Move it" in the detail panel, catches the student who arrives by reading. The first-visit hint points at it once.

What should Superficial and Deep do to the model?

Deep peels the surface layer to a ghost; Surface leaves the model whole.

Rename it "Layer: Surface · Deep" and put it beside the views. Deep drops superficial muscles to 12% opacity and out of hover, keeps deep muscles solid, filters the list to match, and shows "Surface layer peeled · approximate" on the stage. Surface filters the list only. That matches what the labels promise and what a dissection does. Keep "approximate" because the classification is from bounds.

Is the intro banner worth its space?

No.

It costs 152px on every visit for a sentence the brand line already says, and it is the reason the library and the camera buttons are below the fold. Its one useful control, the quiz button, moves to the top nav. If you want a welcome, it belongs inside the right panel's empty state, once, and it should be three actions, not a headline.

What is the first thing a new visitor should see and do?

See the model, large, on a dark ground. Do: click a muscle.

Everything else in the app flows from a selection: facts, attachments, joints, playlists, quizzes. So the first frame should make the model the obvious thing to touch: the biggest object on the page, high contrast, a hover name, and one line on the stage, "Click any muscle". The right panel's empty state offers the three other doors (a study set, a quiz, a fascial line) for the student who arrived with a task. A subtle one-time highlight of, say, the deltoid on first load, with a name label, would show what clicking does before anyone clicks; skip it under reduced motion.

08 **Issues to file**

One per row, titled so they can go straight onto the tracker. Priority and cost as above. I have not filed them; say the word and I will.

#	ISSUE TITLE	PRIORITY	COST
1	Stage: dark slate ground, remove dot grid, light stage text tokens	P0	cheap
2	Remove intro banner and start-panel headline; workspace fills the viewport; Quiz in top nav	P0	cheap
3	Filters: only active filters filled; Region and Joint as selects; list within 260px of panel top	P0	cheap
4	Stage toolbar with "Move a joint" menu; "Move the ..." link in Crosses row; rename Animate	P0	cheap
5	Layer: Deep ghosts superficial muscles on the model with a stage caption	P0	cheap-medium
6	Selection colour #2ac7e0 with bright rim; pins avoid origin/insertion hues; motion cables amber/blue	P0	cheap
7	Panels: "Find a structure" / "Selected" headings; attachments chip under the name; "Start here" empty state	P1	cheap
8	One collapsible dock for quiz, playlist and motion cards; gizmo and zoom to a corner stack	P1	medium
9	Type scale with a 12px floor; fix the seven failing text colours	P1	cheap
10	Reduced motion: hide Play, explain, keep the scrubber; tour says auto-advance is off	P1	cheap
11	Phone: full-height stage, search pill, half-height Find sheet, bottom sheet for details and cards	P1	medium
12	Caveats as always-visible chips that expand; legend caveat in the caption band at 12px	P1	cheap
13	Copy pass (labels, suffixes, badge gloss inline)	P2	cheap
14	Three anchored first-visit hints replace the guide as onboarding	P2	medium
15	Ghost occluders when a list, search, playlist or quiz selection is hidden from the camera	P2	engineering
16	CI: axe colour-contrast in headless Chrome at laptop and phone widths	P2	cheap
17	Search: Enter selects the first match; "1 match" strip under the box	P2	cheap

09 Method notes

How the numbers were taken

Reviewed for the Form team, 7 September 2026. Nothing in this document changes the model, the data or the sources; every recommendation is colour, layout, copy, defaults or placement of existing actions, except issues 8, 11, 14 and 15, which are marked medium or engineering.